Bioinformatics algorithms

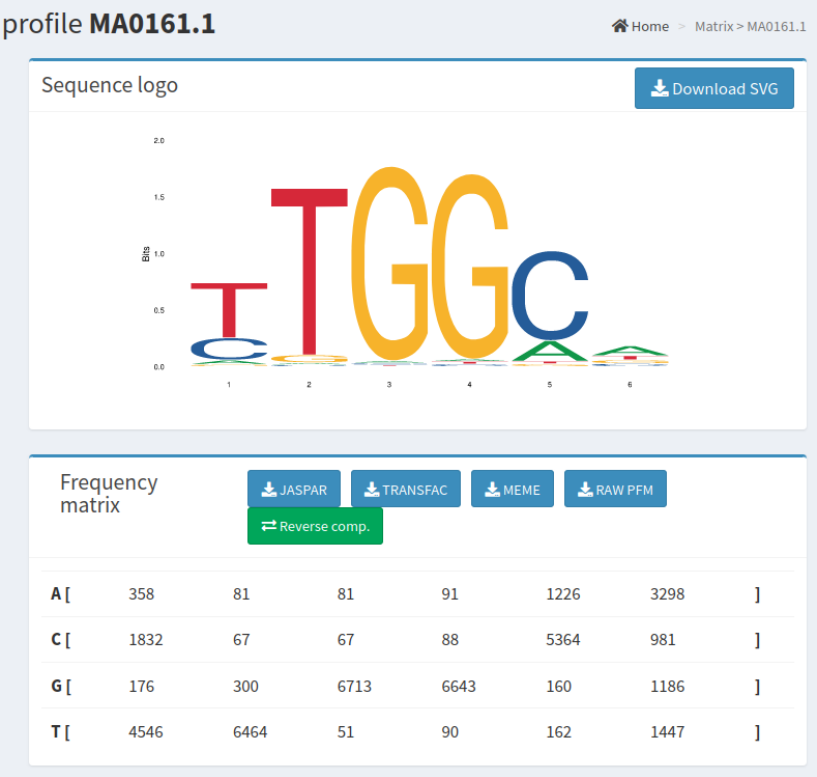
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Sequence modeling and Hidden Markov Models – exercises (**with solutions**)

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1. **Alignment profile/motif**

   Consider the following motif (equivalent to an alignment profile) of the transcription factor *NFIC*:



   Note: the sequence logo and profile data are from the JASPAR database (with matrix/profile ID MA0161.1) and can be found at `https://jaspar.genereg.net/matrix/MA0161.1/`

   Answer the following questions:

   (a) This profile is given as a "Position Frequency Matrix" (PFM) which reports for every position in the alignment the number of times a specific letter was observed at that position (i.e., how many of the aligned strings contain that letter at that position). Transform the PFM into a probability matrix as we have seen it in the previous lecture:

	Pos.1	Pos.2	Pos.3	Pos.4	Pos.5	Pos.6
A	$f_{A,1}$	$f_{A,2}$	$f_{A,3}$	$f_{A,4}$	$f_{A,5}$	$f_{A,6}$
C	$f_{C,1}$	$f_{C,2}$	$f_{C,3}$	$f_{C,4}$	$f_{C,5}$	$f_{C,6}$
G	$f_{G,1}$	$f_{G,2}$	$f_{G,3}$	$f_{G,4}$	$f_{G,5}$	$f_{G,6}$
T	$f_{T,1}$	$f_{T,2}$	$f_{T,3}$	$f_{T,4}$	$f_{T,5}$	$f_{T,6}$

   Solution

   Simply divide the elements in each column (number observations for a specific letter at the given position) by their sum (total number of observations at that position)!

	Pos.1	Pos.2	Pos.3	Pos.4	Pos.5	Pos.6
A	$f_{A,1} = 0.052$	$f_{A,2} = 0.012$	$f_{A,3} = 0.012$	$f_{A,4} = 0.013$	$f_{A,5} = 0.177$	$f_{A,6} = 0.477$
C	$f_{C,1} = 0.265$	$f_{C,2} = 0.010$	$f_{C,3} = 0.010$	$f_{C,4} = 0.013$	$f_{C,5} = 0.776$	$f_{C,6} = 0.142$
G	$f_{G,1} = 0.025$	$f_{G,2} = 0.043$	$f_{G,3} = 0.971$	$f_{G,4} = 0.961$	$f_{G,5} = 0.023$	$f_{G,6} = 0.172$
T	$f_{T,1} = 0.658$	$f_{T,2} = 0.935$	$f_{T,3} = 0.007$	$f_{T,4} = 0.013$	$f_{T,5} = 0.023$	$f_{T,6} = 0.209$

   (Note: the columns need to sum up to 1 = 100%, but this is not guaranteed when rounding the values; for position 5 the sum of the values in the table is 0.999)

(b) What is the "consensus" sequence of this motif?

Solution
For each position the most frequent letter: `TTGGCA`

(c) Why does the profile contain no row for the gap ("-") symbol?

Solution
This is derived from a local multiple sequence alignment where all aligned substrings are considered to have the same length, so there simply are no gaps!

(d) Consider the following two DNA sequences of the same length:
Sequence 1: `TGGGCA`
Sequence 2: `CTGGCT`

1. Which of the two sequences is more similar to the consensus sequence using the Hamming distance?
2. Which of the two sequences fits the motif better? (Any idea of how to compute this? No need to calculate a final solution, just show how you would be able to obtain it ...)

Solution

1. Sequence 1 is closest to the consenus, only the second base is different (`G` instead of `T`); sequence 2 appears less similar, having two changes (`C` instead `T` in the first position, `T` instead of `A` in the last position)
2. We can compute the overall probability of observing these sequences by multiplying the probabilities/fractions of the individual nucleotides:

   For sequence 1 (`TGGGCA`):
   $P(\texttt{TGGGCA}) = \texttt{f}_{T,1} \times \texttt{f}_{G,2} \times \texttt{f}_{G,3} \times \texttt{f}_{G,4} \times \texttt{f}_{C,5} \times \texttt{f}_{A,6} =$
   $= 0.658 \times 0.043 \times 0.971 \times 0.961 \times 0.776 \times 0.477 = 9.77 \times 10^{-3}$

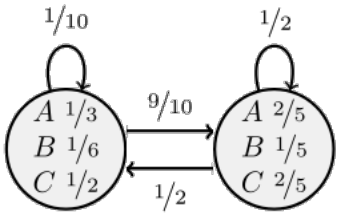
   For sequence 2 (`CTGGCT`):
   $P(\texttt{CTGGCT}) = \texttt{f}_{C,1} \times \texttt{f}_{T,2} \times \texttt{f}_{G,3} \times \texttt{f}_{G,4} \times \texttt{f}_{C,5} \times \texttt{f}_{T,6} =$
   $= 0.265 \times 0.935 \times 0.971 \times 0.961 \times 0.776 \times 0.209 = 3.75 \times 10^{-2}$

   Thus, sequence 2 is actually nearly $3.75 \times 10^{-2} / 9.77 \times 10^{-3} \approx 4$ times more likely than sequence 1!

   (For the exam, developing the formula for $P(\texttt{TGGGCA})$ and $P(\texttt{CTGGCT})$ and inserting the correct frequencies/probabilities would have been enough to show knowledge of the procedure; no need to bring a calculator!)

2. **Hidden Markov Model (HMM)**

Consider the following graph representing a very simple HMM:



Let's call the two hidden states "L" (left) and "R" (right). The observable symbols which can be emitted by both states are "A", "B" and "C".

The transision probabilites a_{LL}, a_{LR}, a_{RR} and a_{RL}, as well as the emission probabilities $e_L(A)$, $e_L(B)$, $e_L(C)$, $e_R(A)$, $e_R(B)$, and $e_R(C)$, can be obtained from the figure.

Assume the initial state probabilies to be: $P_{start}(L) = 2/10$ and $P_{start}(R) = 8/10$.

(a) Calculate the probability of having the following given hidden path π and observed sequence x:

$$\pi = \texttt{R R L L R L R}$$
$$x = \texttt{B A C A C A A}$$

(No need to calculate the final result. Just compose the equation out of the individual probabilities to show how to compute the final probability.)

Solution

$P(x, \pi) = P_{start}(R) \times e_R(B) \times a_{RR} \times e_R(A) \times a_{RL} \times e_L(C) \times a_{LL} \times e_L(A) \times a_{LR} \times e_R(C) \times a_{RL} \times e_L(A) \times a_{LR} \times e_R(A)$

$P(x, \pi) = 8/10 \times 1/5 \times 1/2 \times 2/5 \times 1/2 \times 1/2 \times 1/10 \times 1/3 \times 9/10 \times 2/5 \times 1/2 \times 1/3 \times 9/10 \times 2/5$

$P(x, \pi) = 5.76 \times 10^{-6}$

(For the exam, developing the first two lines would have been enough to show knowledge of the procedure; no need to bring a calculator!)

(b) How would you compute the overall probability to obtain the sequence "CAB" from this HMM? No need to calculate the final result, just show the procedure!

Solution

We can obtain the sequence x="CAB" from all possible hidden paths of length 3 (here: $2^3 = 8$ paths), so we have to sum the probabilities of all these possibilies:

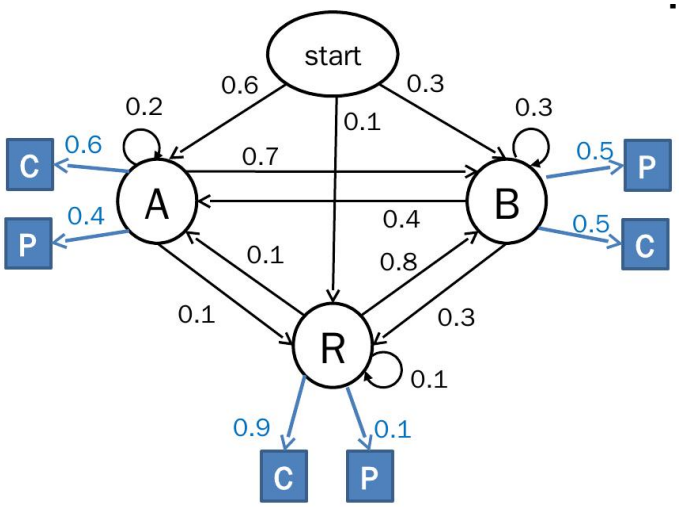
$P(x) = \sum_\pi P(x, \pi) =$
$= P(x, LLL) + P(x, LLR) + P(x, LRL) + P(x, LRR) +$
$+ P(x, RLL) + P(x, RLR) + P(x, RRL) + P(x, RRR) = \ldots$

Much better (and much more efficient): the dynamic programming approach we have seen in the lecture!

3. **Viterbi algorithm and most likely path**

(Note: this exercise serves to let you better understand how the Viterbi algorithm works, but the procedure is too long as an exercise for the exam; you should however know what purpose this algorithm serves and how it works!)

Consider the following HMM:



The HMM shown in the figure represents three possible hidden states ("A", "B" and "R") and two observable symbols ("C" and "P"). The **black** arrows show the initial state probabilities (from "start") and the transition probabilities between the three states. The **blue** arrows show the emission probabilities associated with the individual states.

What is the most likely hidden path to have produced the following sequence of observed symbols?

P C C P C P

Use the Viterbi algorithm to fill the following matrix and find the solution.

Step i	1	2	3	4	5	6
Emission	P	C	C	P	C	P
A	0.24	0.036 (B at i=1)	0.020 (B at i=2)	2.6*10^{-3} (B at i=3)	2.2*10^{-3} (B at i=4)	2.2*10^{-4} (B at i=5)
B	0.15	0.084 (A at i=1)	0.016 (R at i=2)	9.1*10^{-3} (R at i=3)	1.4*10^{-3} (B at i=4)	**9.8*10**$^{-4}$ (R at i=5)
R	0.01	0.041 (B at i=1)	0.023 (B at i=2)	4.9*10^{-4} (B at i=3)	2.4*10^{-3} (B at i=4)	4.1*10^{-5} (B at i=5)

(Probabilities are rounded; see below for the correct ones)

Solution

For the single columns:

1. Emission of "P" at position i=1:
   - Probability of state "A" (initial state * emission prob.): $V_A(1) = p(A) \times e_A(P) = 0.6 \times 0.4 = 0.24$
   - Probability of state "B" (initial state * emission prob.): $V_B(1) = p(B) \times e_B(P) = 0.3 \times 0.5 = 0.15$
   - Probability of state "R" (initial state * emission prob.): $V_R(1) = p(R) \times e_R(P) = 0.1 \times 0.1 = 0.01$
2. Emission of "C" at position i=2:
   - Probability of state "A":
     - Coming from "A" at i=1: $e_A(C) \times a_{AA} \times V_A(1) = 0.6 \times 0.2 \times 0.24 = 0.0288$
     - Coming from "B" at i=1: $e_A(C) \times a_{BA} \times V_B(1) = 0.6 \times 0.4 \times 0.15 = 0.036$ (best choice!)
     - Coming from "R" at i=1: $e_A(C) \times a_{RA} \times V_R(1) = 0.6 \times 0.1 \times 0.01 = 0.0006$

     So we have $V_A(2) = e_A(C) \times \max_j a_{jA} \times V_j(1) = 0.036$! Move (for traceback): B→A
   - Probability of state "B":
     - Coming from "A" at i=1: $e_B(C) \times a_{AB} \times V_A(1) = 0.5 \times 0.7 \times 0.24 = 0.084$ (best choice!)
     - Coming from "B" at i=1: $e_B(C) \times a_{BB} \times V_B(1) = 0.5 \times 0.3 \times 0.15 = 0.0225$
     - Coming from "R" at i=1: $e_B(C) \times a_{RB} \times V_R(1) = 0.5 \times 0.8 \times 0.01 = 0.004$

     So we have $V_B(2) = e_B(C) \times \max_j a_{jB} \times V_j(1) = 0.084$! Move (for traceback): A→B
   - Probability of state "R":
     - Coming from "A" at i=1: $e_R(C) \times a_{AR} \times V_A(1) = 0.9 \times 0.1 \times 0.24 = 0.0216$
     - Coming from "B" at i=1: $e_R(C) \times a_{BR} \times V_B(1) = 0.9 \times 0.3 \times 0.15 = 0.0405$ (best choice!)
     - Coming from "R" at i=1: $e_R(C) \times a_{RR} \times V_R(1) = 0.9 \times 0.1 \times 0.01 = 0.0009$

     So we have $V_R(2) = e_R(C) \times \max_j a_{jR} \times V_j(1) = 0.0405$! Move (for traceback): B→R
3. Emission of "C" at position i=3:
   - Probability of state "A":
     - Coming from "A" at i=2: $e_A(C) \times a_{AA} \times V_A(2) = 0.6 \times 0.2 \times 0.036 = 0.00432$
     - Coming from "B" at i=2: $e_A(C) \times a_{BA} \times V_B(2) = 0.6 \times 0.4 \times 0.084 = 0.02016$ (best choice!)
     - Coming from "R" at i=2: $e_A(C) \times a_{RA} \times V_R(2) = 0.6 \times 0.1 \times 0.0405 = 0.00243$

     So we have $V_A(3) = e_A(C) \times \max_j a_{jA} \times V_j(2) = 0.02016$! Move (for traceback): B→A
   - Probability of state "B":
     - Coming from "A" at i=2: $e_B(C) \times a_{AB} \times V_A(2) = 0.5 \times 0.7 \times 0.036 = 0.0126$
     - Coming from "B" at i=2: $e_B(C) \times a_{BB} \times V_B(2) = 0.5 \times 0.3 \times 0.084 = 0.0126$
     - Coming from "R" at i=2: $e_B(C) \times a_{RB} \times V_R(2) = 0.5 \times 0.8 \times 0.0405 = 0.0162$ (best choice!)

So we have $V_B(3) = e_B(C) \times \max_j a_{jB} \times V_j(2) = 0.0162$! Move (for traceback): R→B

- Probability of state "R":
  - Coming from "A" at i=2: $e_R(C) \times a_{AR} \times V_A(2) = 0.9 \times 0.1 \times 0.036 = 0.00324$
  - Coming from "B" at i=2: $e_R(C) \times a_{BR} \times V_B(2) = 0.9 \times 0.3 \times 0.084 = 0.02268$ (best choice!)
  - Coming from "R" at i=2: $e_R(C) \times a_{RR} \times V_R(2) = 0.9 \times 0.1 \times 0.0405 = 0.003645$

  So we have $V_R(3) = e_R(C) \times \max_j a_{jR} \times V_j(2) = 0.02268$! Move (for traceback): B→R

4. Emission of "P" at position i=4:
   - Probability of state "A":
     - Coming from "A" at i=3: $e_A(P) \times a_{AA} \times V_A(3) = 0.4 \times 0.2 \times 0.02016 = 0.0016128$
     - Coming from "B" at i=3: $e_A(P) \times a_{BA} \times V_B(3) = 0.4 \times 0.4 \times 0.0162 = 0.002592$ (best choice!)
     - Coming from "R" at i=3: $e_A(P) \times a_{RA} \times V_R(3) = 0.4 \times 0.1 \times 0.02268 = 0.0009072$

     So we have $V_A(4) = e_A(P) \times \max_j a_{jA} \times V_j(3) = 0.002592$! Move (for traceback): B→A
   - Probability of state "B":
     - Coming from "A" at i=3: $e_B(P) \times a_{AB} \times V_A(3) = 0.5 \times 0.7 \times 0.02016 = 0.007056$
     - Coming from "B" at i=3: $e_B(P) \times a_{BB} \times V_B(3) = 0.5 \times 0.3 \times 0.0162 = 0.00243$
     - Coming from "R" at i=3: $e_B(P) \times a_{RB} \times V_R(3) = 0.5 \times 0.8 \times 0.02268 = 0.009072$ (best choice!)

     So we have $V_B(4) = e_B(P) \times \max_j a_{jB} \times V_j(3) = 0.009072$! Move (for traceback): R→B
   - Probability of state "R":
     - Coming from "A" at i=3: $e_R(P) \times a_{AR} \times V_A(3) = 0.1 \times 0.1 \times 0.02016 = 0.0002016$
     - Coming from "B" at i=3: $e_R(P) \times a_{BR} \times V_B(3) = 0.1 \times 0.3 \times 0.0162 = 0.000486$ (best choice!)
     - Coming from "R" at i=3: $e_R(P) \times a_{RR} \times V_R(3) = 0.1 \times 0.1 \times 0.02268 = 0.0002268$

     So we have $V_R(4) = e_R(P) \times \max_j a_{jR} \times V_j(3) = 0.000486$! Move (for traceback): B→R
5. Emission of "C" at position i=5:
   - Probability of state "A":
     - Coming from "A" at i=4: $e_A(C) \times a_{AA} \times V_A(4) = 0.6 \times 0.2 \times 0.002592 = 0.00031104$
     - Coming from "B" at i=4: $e_A(C) \times a_{BA} \times V_B(4) = 0.6 \times 0.4 \times 0.009072 = 0.00217728$ (best choice!)
     - Coming from "R" at i=4: $e_A(C) \times a_{RA} \times V_R(4) = 0.6 \times 0.1 \times 0.000486 = 2.916 * 10^{-5}$

     So we have $V_A(5) = e_A(C) \times \max_j a_{jA} \times V_j(4) = 0.00217728$! Move (for traceback): B→A
   - Probability of state "B":
     - Coming from "A" at i=4: $e_B(C) \times a_{AB} \times V_A(4) = 0.5 \times 0.7 \times 0.002592 = 0.0009072$
     - Coming from "B" at i=4: $e_B(C) \times a_{BB} \times V_B(4) = 0.5 \times 0.3 \times 0.009072 = 0.0013608$ (best choice!)
     - Coming from "R" at i=4: $e_B(C) \times a_{RB} \times V_R(4) = 0.5 \times 0.8 \times 0.000486 = 0.0001944$

     So we have $V_B(5) = e_B(C) \times \max_j a_{jB} \times V_j(4) = 0.0013608$! Move (for traceback): B→B
   - Probability of state "R":
     - Coming from "A" at i=4: $e_R(C) \times a_{AR} \times V_A(4) = 0.9 \times 0.1 \times 0.002592 = 0.00023328$
     - Coming from "B" at i=4: $e_R(C) \times a_{BR} \times V_B(4) = 0.9 \times 0.3 \times 0.009072 = 0.00244944$ (best choice!)
     - Coming from "R" at i=4: $e_R(C) \times a_{RR} \times V_R(4) = 0.9 \times 0.1 \times 0.000486 = 4.374 * 10^{-5}$

     So we have $V_R(5) = e_R(C) \times \max_j a_{jR} \times V_j(4) = 0.00244944$! Move (for traceback): B→R
6. Emission of "P" at position i=6:
   - Probability of state "A":
     - Coming from "A" at i=5: $e_A(P) \times a_{AA} \times V_A(5) = 0.4 \times 0.2 \times 0.00217728 = 0.0001741824$
     - Coming from "B" at i=5: $e_A(P) \times a_{BA} \times V_B(5) = 0.4 \times 0.4 \times 0.0013608 = 0.000217728$ (best choice!)

- Coming from "R" at i=5: $e_A(P) \times a_{RA} \times V_R(5) = 0.4 \times 0.1 \times 0.00244944 = 9.79776 * 10^{-5}$

So we have $V_A(6) = e_A(P) \times \max_j a_{jA} \times V_j(5) = 0.000217728$! Move (for traceback): B→A

- Probability of state "B":
  - Coming from "A" at i=5: $e_B(P) \times a_{AB} \times V_A(5) = 0.5 \times 0.7 \times 0.00217728 = 0.000762048$
  - Coming from "B" at i=5: $e_B(P) \times a_{BB} \times V_B(5) = 0.5 \times 0.3 \times 0.0013608 = 0.00020412$
  - Coming from "R" at i=5: $e_B(P) \times a_{RB} \times V_R(5) = 0.5 \times 0.8 \times 0.00244944 = 0.000979776$ (best choice!)

  So we have $V_B(6) = e_B(P) \times \max_j a_{jB} \times V_j(5) = 0.000979776$! Move (for traceback): R→B
- Probability of state "R":
  - Coming from "A" at i=5: $e_R(P) \times a_{AR} \times V_A(5) = 0.1 \times 0.1 \times 0.00217728 = 2.17728 * 10^{-5}$
  - Coming from "B" at i=5: $e_R(P) \times a_{BR} \times V_B(5) = 0.1 \times 0.3 \times 0.0013608 = 4.0824 * 10^{-5}$ (best choice!)
  - Coming from "R" at i=5: $e_R(P) \times a_{RR} \times V_R(5) = 0.1 \times 0.1 \times 0.00244944 = 2.44944 * 10^{-5}$

  So we have $V_R(6) = e_R(P) \times \max_j a_{jR} \times V_j(5) = 4.0824 * 10^{-5}$! Move (for traceback): B→R

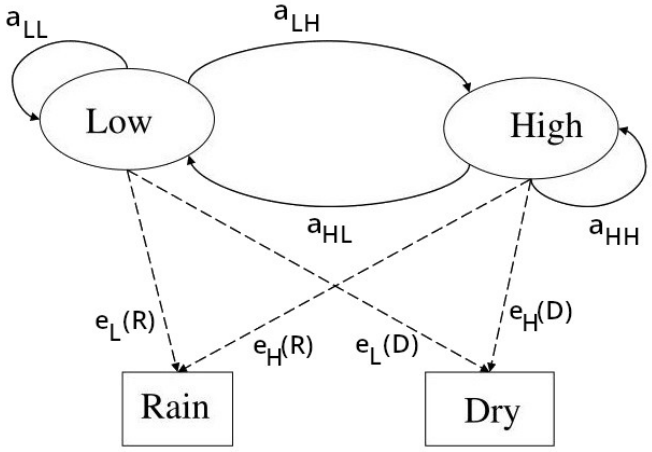
Thus, the most likely hidden path, which ends in state B (best probability of 9.8*10^{-4}), is the following (see traceback):

A	→	B	→	R	→	B	→	R	→	B
↓		↓		↓		↓		↓		↓
P		C		C		P		C		P

Note: the state R has the highest probability (0.9) to emit a "C". So why is the second hidden state a B instead? Because the transition probability directly from A to R is very low (0.1). So the best (most likely) way to get two "C"s at positions 2 and 3 is to first move from A to B and then to R!

4. **Supervised learning of HMM parameters**

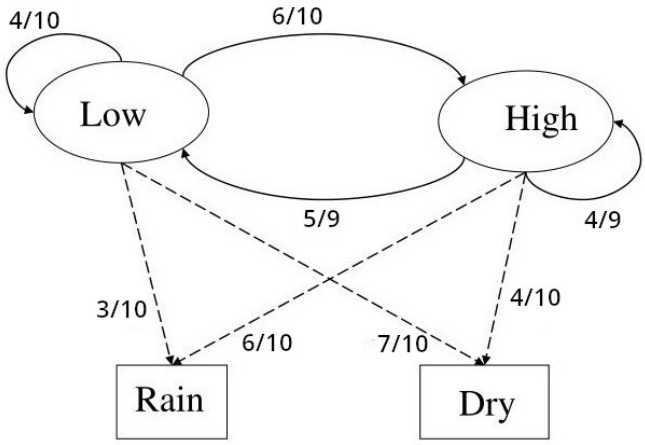
Consider the following HMM:



(a) Determine the maximum likelihood parameters of the HMM from the following labeled training data:

π = L H H H L L H H H L H L H L L L H L L H
x = R D D R D R R R D D R R R D D D R D D D

Solution



(b) Are there some parameters that cannot be easily estimated from this training data?

Solution

We cannot estimate the initial state probabilities because we have only one single example (only one start). We would need multiple starts to estimate initial state probabilities (= frequencies at which the two states are selected at the first step).